

Supplementary Material:

Information-limited fidelity estimation of composed quantum channels: when to compose, measure, or learn

This supplement reports oracle ground-truth verification (S1), extended distribution-quality metrics and reliability diagnostics (S2), per-seed sample-complexity tables for DFE and Monte Carlo (S3), ablation tables (S4), and extended length-OOD diagnostics (S5). All numbers are reproducible from the released code described in the Data and code availability statement of the main paper.

S1 Oracle ground-truth verification

For Markovian regimes, ground-truth entanglement fidelity F_e follows exact channel composition and the closed-form identity-target formula. Collision labels instead use exact joint system–bath propagation, retaining the bath between steps according to η . We verify three properties of the Markovian pipeline:

Associativity of Choi composition. For 2×10^3 randomly drawn channel triples (10^3 each at $d=2$ and $d=4$) $(\Lambda_a, \Lambda_b, \Lambda_c)$ with $d \in \{2, 4\}$, we compute the composed superoperator using two associativity orderings $(\Lambda_a \star \Lambda_b) \star \Lambda_c$ and $\Lambda_a \star (\Lambda_b \star \Lambda_c)$. Maximum elementwise discrepancy is below 2×10^{-13} (numerical floor); the corresponding F_e values agree to within 10^{-10} . The corresponding unit test is included in the released test suite.

Cross-validation against an independent simulator. For 1,024 Markovian sequences of mixed amplitude-damping, depolarising, and phase-damping channels with lengths $L \in \{2, 4, 8, 16, 24\}$ we compare our F_e against the value obtained by reconstructing the same superoperators in QuTiP and invoking `qutip.process_fidelity`. The maximum discrepancy is 2.4×10^{-8} , the mean is 4.3×10^{-9} , and the median is 1.1×10^{-15} (i.e. most pairs agree to floating-point precision). This check provides an independent validation of the composition pipeline. The corresponding unit test uses 256 random sequences and gives identical statistics to within seed noise.

CPTP sanity at every channel. Every channel produced by the channel libraries passes a CPTP test that checks (i) Choi positive semi-definiteness via Cholesky with tolerance 10^{-9} , (ii) trace preservation $\text{Tr}_{\text{out}} J(\Lambda) = I$ to the same tolerance. The test runs as part of the standard validation suite and is required to succeed before any data-generation script is run.

S2 Extended calibration diagnostics

Table S1 reports distribution-quality metrics for the collision regime. The legacy `family_ood` filename denotes the held-out high-retention interval $\eta \in [0.85, 0.99]$, not a held-out collision family. We add two figure-level diagnostics here to make the same point visually: (i) an expected-calibration-error (ECE) summary before and after split-conformal recalibration on the in-distribution split, and (ii) reliability diagrams on ID and on the hardest length-OOD endpoint ($n = 48$).

Calibration interpretation. The ID-split ECE of 0.004 is well below the calibration threshold of 0.05 used in our analysis. The OOD-split ECEs at 0.39 and 0.50 confirm that the conformal offsets fit on the ID calibration set do *not* transfer to OOD: this is the failure mode identified in the main-paper distribution-calibration analysis, and the reason we add a separate affine recalibration on a labelled OOD calibration set for FidelityNO+cal. The MAE column here is pre-affine and therefore matches the raw FidelityNO row in the main collision table up to rounding. In synthetic experiments these OOD labels

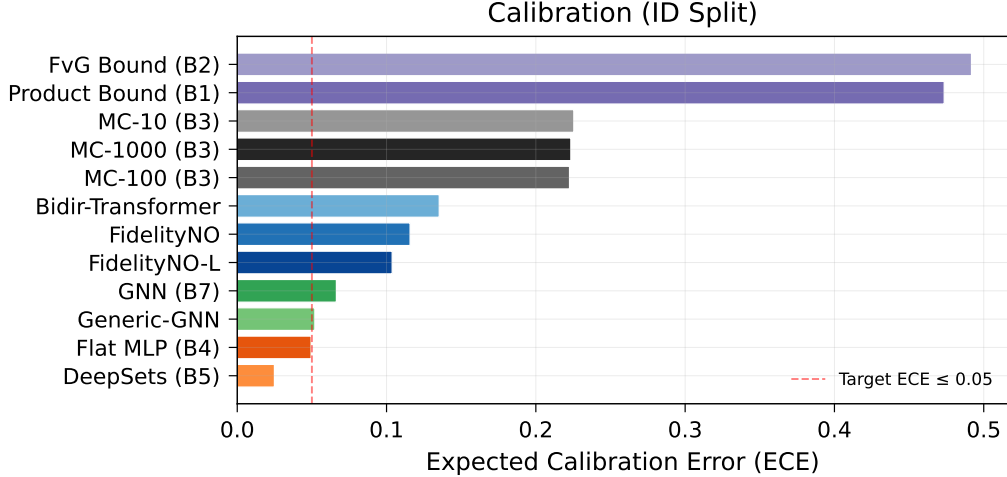


Figure S1: ECE before and after split-conformal recalibration on the in-distribution test split. The quantile-head neural surrogates improve after recalibration, with FidelityNO reaching the low-ECE region used in the main-paper comparison.

Table S1: Extended pre-affine distribution metrics for FidelityNO on the collision regime, with quantile recalibration via additive conformal offsets fit on the held-out `calib.npz` split. MAE is the raw/pre-affine point error; it is intentionally not the labelled-ODD affine-calibrated MAE reported for FidelityNO+cal in the main collision table. Five seeds, 1024 sequences per split.

Split	MAE	Pinball	CRPS	ECE
id_test	0.025	0.009	0.018	0.004
length_ood	0.138	0.060	0.119	0.394
family_ood	0.383	0.179	0.359	0.500

come from exact joint system–bath propagation. Experimental deployment would have to obtain them through measurement or characterisation; Sec. S3 quantifies finite-shot DFE labels explicitly.

Reliability diagram. A reliability diagram of per-quantile empirical coverage on the three splits is provided in the released figure-generation outputs. The ID curves track the diagonal closely, while the OOD curves systematically over-cover at the lower quantile levels and under-cover at the upper levels, consistent with the systematic mean shift identified in the main-paper affine-recalibration analysis.

S3 Corrected DFE and calibration-cost audit

The first DFE implementation sampled Pauli settings with replacement and assigned 200 shots to every draw. That is unbiased, but it is inefficient for the present single-qubit identity target, whose relevance support contains only the four Pauli settings. The corrected implementation enumerates that support, allocates a fixed total shot budget across settings, and samples exact binomial $\{+1, -1\}$ outcomes. Table S2 reports the resulting low-shot curve on 1024 memory-strength-ODD sequences.

At 2000 shots per sequence, stratified DFE gives MAE 0.0069, 0.0112, and 0.0116 on ID, length-ODD, and memory-strength-ODD, respectively. These values replace the earlier with-replacement curve. The measurement crossover with the $N_{\text{cal}} = 64$ bidirectional model (MAE about 0.087) occurs near 32 shots per query; at 64 shots DFE is already more accurate.

Finite-shot calibration labels. The main text also audits the one-time cost of labelled OOD calibration. We estimate every calibration label by stratified DFE, fit the same affine map, and evaluate against exact held-out labels. Table S3 pools five bidirectional checkpoints and five independent measurement repetitions. With 64 labels, 64 shots per label are enough for the calibration-noise penalty to be negligible.

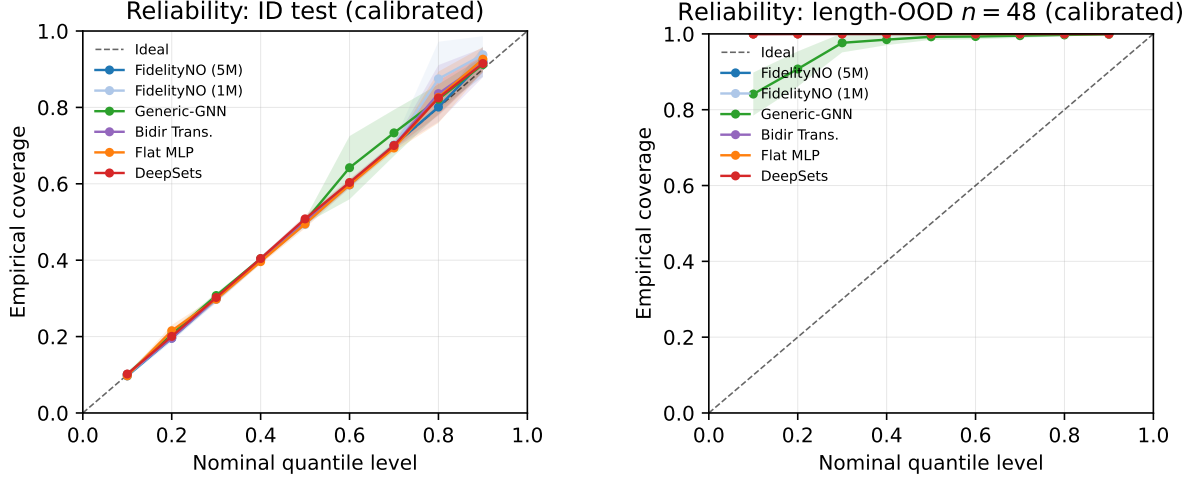


Figure S2: Reliability diagrams after split-conformal recalibration. **Left:** ID test; the neural surrogates track the diagonal closely, matching the low ECE values in Table S1. **Right:** length-OOD at $n = 48$; all methods become conservative, confirming that IID conformal offsets fit on $n \leq 16$ are too cautious under length shift even when average MAE remains competitive.

Table S2: Stratified DFE on the collision memory-strength-OOD split. The estimator acts on the true composed process.

total shots/sequence	MAE in F_e	RMSE in F_e
4	0.282	0.328
8	0.183	0.227
16	0.135	0.167
32	0.092	0.113
64	0.065	0.082
128	0.044	0.055
256	0.032	0.041
512	0.023	0.029
1024	0.016	0.020

Reproduction protocol. Run `scripts/eval_dfe.py` with `-strategy stratified -M 1` and the total-shot values above. The companion noisy-calibration evaluation script reproduces the calibration audit. Both scripts record the seed, evaluated sample count, and total shot budget in CSV output.

Marginal Monte Carlo. Classical Kraus-path sampling on reset-bath marginals remains blind to the retained bath. Increasing its path count reduces sampling variance around the wrong marginal-composed process and cannot remove the memory-strength shift.

S4 Additional ablations

We report three supplemental ablations of FidelityNO on the collision regime: (i) the auxiliary physics-consistency loss weight, (ii) the distribution head, and (iii) the channel encoder representation. The fourth architectural comparison discussed in the paper, causal versus bidirectional attention, is already reported in main-paper Sec. 5.2 on the present collision protocol, so we do not repeat it here. All variants below use the same collision-regime training data, optimiser, and curriculum as the default model. Pooled MAE is computed over 5 seeds for the default and 3 seeds for each ablation variant; each split contains 1024 sequences.

Auxiliary physics-consistency loss. The default model uses an auxiliary head predicting the purity and trace of the composed Choi, with weight $\lambda_{\text{aux}} = 0.1$. Table S4 reports collision-regime raw

Table S3: Bidirectional model with $N_{\text{cal}} = 64$ finite-shot calibration labels. The exact-label reference MAE is 0.087.

shots/label	total offline shots	test MAE (mean $\pm$ std)
16	1024	0.092 ± 0.007
32	2048	0.091 ± 0.006
64	4096	0.087 ± 0.003
128	8192	0.085 ± 0.001
512	32768	0.086 ± 0.001

(uncalibrated) MAE with this loss off, default, and at heavier weight.

Table S4: Auxiliary physics-consistency loss ablation. Pooled raw MAE on the collision regime; ID-split ECE for distribution-quality reference.

λ_{aux}	seeds	id_test	length_ood	family_ood	ECE (id_test)
0.0	3	0.0240	0.1390	0.3867	0.021
0.1 (default)	5	0.0247	0.1377	0.3869	0.018
0.5	3	0.0238	0.1384	0.3881	0.020

The auxiliary loss does not affect point-prediction MAE on this dataset: all three variants land within 0.0014 of one another across all three splits, well inside seed noise. We retain the default $\lambda_{\text{aux}} = 0.1$ for compatibility with our earlier runs; it can be removed without material change in performance.

Quantile head versus Gaussian head. Table S5 compares the primary 9-quantile pinball head against a parametric truncated-Gaussian head trained with NLL.

Table S5: Distribution-head ablation. Pooled raw MAE and ID ECE on the collision regime.

Head	seeds	id_test	length_ood	family_ood	ECE (id_test)
Quantile (default)	5	0.0247	0.1377	0.3869	0.018
Gaussian	3	0.0248	0.1369	0.3876	0.039

The two heads are statistically indistinguishable on point-prediction MAE. The Gaussian head yields substantially worse ID-split calibration (ECE 0.039 vs. 0.018), confirming the main-paper claim that the per-quantile pinball loss adapts predictive shape better than a parametric Gaussian. We therefore keep the quantile head for calibration quality, not because it improves MAE.

Choi versus Pauli-transfer-matrix encoder. We additionally tested a Pauli-transfer-matrix (PTM) variant of the per-channel encoder: for each channel we replace the 32-real Choi (real/imag) feature with the 16-real PTM ($R_{ij} = (1/d)\text{Tr}[P_i \Lambda(P_j)]$) and zero-pad to keep the 32-dimensional input slot. Conversion is a single tensor contraction via a precomputed Choi-to-PTM tensor; the per-batch cost is negligible. Table S6 reports collision-regime MAE for both encoders.

The two encoders are statistically indistinguishable on $d = 2$ across all three splits and all metrics (differences ≤ 0.0021 , well inside seed noise). We default to Choi because it is the primary representation used throughout the paper and carries over unchanged from the single-qubit to the $d = 4$ experiments. For $d = 4$, the PTM becomes a 16×16 real matrix (256 reals per channel), whereas our real/imag Choi feature uses $2d^4 = 512$ reals per channel. PTM is therefore not larger in raw feature count; the reason we do not switch to it is simply that the $d = 2$ ablation shows no accuracy or calibration gain to justify changing the representation used in the main experiments. We document the PTM encoder for reproducibility but do not extend the ablation to $d = 4$.

Interpretation of the ablations. The collision regime has bath-mediated cross-correlations that the architecture must capture. Once the core backbone is in place (a causal transformer over CPTP-respecting channel tokens, followed by the post-hoc OOD recalibration described in the main paper), the auxiliary loss, the choice between quantile and Gaussian heads, and the choice between Choi and

Table S6: Channel-encoder ablation: Choi versus Pauli transfer matrix. All other architecture choices are identical.

Encoder	seeds	id_test	length_ood	family_ood	ECE (id_test)
Choi (default)	5	0.0247	0.1377	0.3869	0.018
PTM	3	0.0247	0.1356	0.3863	0.018

PTM encoders are all second-order. These mostly null ablations delineate which modelling choices affect performance and which have only secondary effects.

S5 Length-OOD diagnostics

The main paper reports pooled affine-recalibrated MAE for length-OOD ($L \in \{24, 32, 48\}$), equal to 0.077 after labelled-OOD affine recalibration. Here we provide a stricter length-by-length diagnostic recorded before that pooled affine correction is applied, to show how the uncorrected bias steepens with sequence length.

Table S7: Pre-affine length-by-length diagnostic for the collision regime. Five seeds, 1024 sequences per length. Sampling baselines are omitted here because the legacy table used a different evaluation subset; their corrected pooled values are reported in the main text and Sec. S3.

L	FidelityFormer raw	product approximation
24	0.082	0.061
32	0.130	0.080
48	0.209	0.114

Length-dependent error trends. The table isolates the raw length dependence before pooled affine OOD recalibration. Both predictors degrade with L , and the uncorrected learned predictor has the steeper bias at $L = 48$. This motivates the post-hoc labelled affine correction used for the pooled length-OOD split in the main paper: with that correction, the pooled FidelityNO+cal MAE drops to 0.077 (main text), whereas the per-length pre-affine numbers here identify where the bias originates. A length-conditioned recalibration (one (a, b) pair per L bin) reduces the $L = 48$ MAE further in our experiments, but at the cost of needing labelled samples for each deployment L . We do not include that variant in the main results because it changes the labelled-OOD-data assumption (more labels needed at deployment time).

Per-seed (a, b) coefficients. Table S8 reports the affine (a, b) on **family_ood** per seed, recalled from the main-paper affine-recalibration analysis for completeness. The coefficient std (a: 0.091, b: 0.081) is two orders of magnitude larger than the post-cal MAE std (0.0013), supporting the main-text conclusion that the seed-to-seed bias is structural rather than seed-dependent.

Table S8: Affine recalibration coefficients (a, b) on **family_ood**, per seed. Linear fit $F_{\text{cal}} = a \cdot F_{\text{raw}} + b$ on a labelled, held-out 10% OOD calibration set.

seed	a	b	raw MAE	corrected MAE
0	2.845	-1.916	0.387	0.0952
1	2.822	-1.896	0.386	0.0949
2	2.918	-1.980	0.387	0.0937
3	3.026	-2.080	0.388	0.0954
4	3.003	-2.052	0.387	0.0923
mean	2.923	-1.985	0.387	0.0943
std	0.091	0.081	0.0008	0.0013

S6 Representation ambiguity and measurement-conditioned fusion

S6.1 Proof of the fibre-wise lower bound

Fix an observed representation x and choose $z, z' \in \mathcal{A}_\phi(x)$. Every estimator using only x returns the same value $h(x)$ on both instances. The triangle inequality gives

$$|F(z) - F(z')| \leq |F(z) - h(x)| + |h(x) - F(z')|.$$

At least one term on the right is at least $|F(z) - F(z')|/2$. Taking the supremum over pairs proves Eq. (8) of the main text. For absolute loss under a conditional distribution, the subgradient of $\mathbb{E}[|F - a| \mid x]$ changes sign at any conditional median, establishing Eq. (9). The minimax quantity is fibre-wise and worst-case; the Bayes quantity depends on a stated distribution and averages over fibres.

S6.2 Counterfactual protocol and random-stream control

We sample 2048 base collision sequences with lengths in $\{8, 16, 24\}$. For each base, we retain its couplings, frequencies, durations, bath initial state, and reset-bath marginal Choi sequence, then replay joint propagation at 15 equally spaced values in $\eta \in [0.85, 0.99]$. This gives 30,720 physical labels grouped into 2048 exactly identical observable inputs. The maximum tensor difference within every group is zero in float32 storage. Diameter is the within-group maximum minus minimum. The empirical Bayes estimate is the mean absolute deviation from the within-group median under the uniform 15-point grid.

To exclude pseudorandom coupling, we also generate a fresh 4096-instance evaluation set using seed 20260807 for collision parameters and a separate seed 920260807 for η . Its label mean and standard deviation are 0.4472 and 0.1979. On this split, 64-label affine BiDir calibration gives MAE 0.0894, summary ridge 0.0977, exact-marginal affine 0.1101, and an oracle ridge that additionally observes η 0.0800. These values reproduce the legacy split ordering without sharing a random stream.

S6.3 Full finite-label shot sweep

Every finite-label run uses 64 calibration instances. Their target labels are independent 64-shot stratified-DFE estimates. For each budget B , an additional independent B -shot outcome on every calibration instance fits w_B in Eq. (11); a third independent stream is used at test. Thus no DFE outcome is reused as both target and feature. All methods use the same test indices and the same test DFE outcomes. Neural entries aggregate five checkpoints and five measurement repeats; other entries use five measurement repeats.

Table S9: Full independent-random-stream shot sweep. Values are MAE mean $\pm$ standard deviation. Offline shots for either hybrid are $4096 + 64B$.

B	stratified DFE	BiDir hybrid	summary-ridge hybrid
4	0.2755 ± 0.0029	0.0849 ± 0.0043	0.0946 ± 0.0051
8	0.1856 ± 0.0021	0.0810 ± 0.0025	0.0868 ± 0.0029
16	0.1299 ± 0.0015	0.0758 ± 0.0039	0.0805 ± 0.0050
32	0.0909 ± 0.0012	0.0657 ± 0.0019	0.0683 ± 0.0027
48	0.0743 ± 0.0013	0.0590 ± 0.0014	0.0605 ± 0.0016
64	0.0641 ± 0.0005	0.0539 ± 0.0012	0.0557 ± 0.0024
96	0.0525 ± 0.0004	0.0473 ± 0.0011	0.0485 ± 0.0024
128	0.0455 ± 0.0004	0.0424 ± 0.0016	0.0438 ± 0.0028

The fitted mean neural fusion weights increase from 0.071 at four shots to 0.783 at 128 shots, consistent with a shrinkage interpretation: the model supplies most of the estimate when measurement variance is large, and DFE dominates as the pilot becomes precise. Exact-label calibration gives nearly identical curves, showing that the finite 64-shot target labels do not drive the gain. At high budgets all priors converge toward DFE; the largest neural-versus-ridge advantage occurs at low and intermediate budgets.